Predictive Air Quality Modeling in Indian Urban Environments using a Dual-Layered Explainable AI Framework

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Submitted by

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Certificate

This is to certify that the project report entitled **Predictive Air Quality Modeling in Indian Urban Environments using a Dual-Layered Explainable AI Framework** submitted by **Kirti Kumari** to the Central University of Haryana, Mahendergarh, in partial fulfillment for the award of the degree of **B. Tech. in Computer Science and Engineering** is a bonafide record of project work carried out by her under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institution or University for the award of any degree or diploma.

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Declaration

I declare that this internship report titled **"Predictive Air Quality Modeling in Indian Urban Environments using a Dual-Layered Explainable AI Framework"** submitted in partial fulfillment of the degree of **B. Tech. in Computer Science and Engineering** is a record of original work carried out by me under the supervision of **Dr.Sunil Singh** and my internal faculty supervisor **Dr. Nitin Goyal**, and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

During the preparation of this work, I have not used any AI-based tool to write any part of this project report.I take full responsibility for the submitted content including similarity.

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I would like to acknowledge that this project was completed entirely by me and not by someone else.

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Abstract

This work addresses urban air quality informatics by introducing the SHAP-Weighted Adaptive Ensemble (SWAE) architecture,a hybrid framework designed to overcome the limitations of traditional predictive models in handling highly volatile atmospheric data.Standalone deep sequential models frequently suffer from error propagation and distribution mismatches during sharp pollution shocks,while high-performing tree ensembles function as unverified, opaque "black boxes".To eliminate these constraints,the SWAE engine was developed and validated across a six-city subcontinental network (Bengaluru, Chennai, Delhi, Hyderabad, Kolkata, and Mumbai) utilizing historical telemetry spanning 2015 to 2020 from the Central Pollution Control Board (CPCB).Unlike passive systems,the SWAE optimization layer leverages game-theoretic feature attributions computed via SHAP to actively adjust weights between gradient-boosted decision trees and dual-layer Stacked LSTMs.Numeric and predictive cross-validation assure good accuracy,reduced root mean squared error (RMSE) in highly volatile states like Delhi,compressing mean absolute error (MAE)across five out of six places.Crucially,when applied on to non-convergent neural states in complex coastal regions like Chennai—where standalone sequence models collapse then the model shifts entirely to resilient tree arrays to preserve tracking stability.The novelty lies in the execution of a closed-loop dual method validation pipeline which utilizes LIME and SHAP to explicitly report the ensemble's feature rankings and decision-making paths.By validating the results of both explainability methods to each others observation,this framework provides the historic trade-off between performance and interpretability and transparency .In results,this doubly-verified approach provides environmental regulatory bodies with the reliable analysis and results regarding pollution to execute targeted emissions controls and provide public health alerts.

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Chapter 1

Introduction

1.1 Background

Air pollution serves as major concern worldwide.From household smoke to industrial residual pollutants, all contribute to its surge.This challenge is primarily driven by an economy-focused human nature,increased industrialization,urban expansion,and careless human behavior towards nature and climate.The impact of air pollution is severe and multifaceted,impacting species health,environmental ecosystems, and the global climate,termed climate change. Air quality directly affects human life and sustainable development;therefore,it is given a pedestal in sustainable development goals [1].Among the pool of tools to measure air quality, the Air Quality Index (AQI) is the most relied upon and widely used by various governments and organizations.AQI works as a metric that aggregates the key pollutants like PM2.5,PM10,SO2,CO,O3,toluene etc.and indicates the severity of air quality [2].Long term exposure to these pollutants causes premature mortality,skin disease,lung disease,respiratory disease ,and cardiac disease.
According to a statistical report ,India ranks 8th among 131 countries in air pollution[3]; hence, it becomes a challenge that must be addressed rationally with scientific methodologies. Because of the non-linearity in the nature of the data provided for AQI predictions and the computational complexities,the traditional statistical and mathematical models like ARIMA cannot capture the variable air pollutants .Therefore, machine learning techniques like Random Forest,Xgboost, etc. are widely accepted and used for their flexibility in learning from huge unstructured data.These models are generally black boxes,focusing on prediction performance but lack transparency in decision making,which inspires a hybrid approach.

1.1.1 Research Area and Problem Statement

As far as India is concerned,recent literature has seen a shift towards advanced computational methods to address the predictive complexities. With the availability of high data from the Central Pollution Control Board (CPCB),recent works have explored the fusion of machine learning and time-series methods for better precision and accuracy across the country .For example,hybrid deep learning techniques combining multiple models like CNN-LSTM have suc-

cessfully captured the spatial-temporal dependencies,while optimized ensemble structures like Extreme Gradient Boosting (XGBoost) and Random Forests have consistently achieved good predictive metrices on multi-pollutant datasets [4].However,these models suffer a limitation of heavily prioritizing predictions at the expense of model-transparency [4].Because of the black box nature of these predictive models,environmental scientists and regulatory authorities face lack of interpretability, which causes bottlenecks in real world policymaking ,health alerts,and regulatory compliance, etc.. Consequently, Explainable Artificial Intelligence (XAI) has emerged as an inseparable paradigm in modern environmental data science.Model-agnostic explanation techniques like SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME)—are building trust by transforming these black box models through interpretability.[5]

A critical examination of contemporary subcontinental literature highlights significant computational progress,yet prominent methodological limitations remain unaddressed.For instance, while Natarajan et al. (2024) achieved high statistical accuracy using tree-based ensembles,their framework completely omitted sequential deep learning pipelines and post-hoc interpretability layers.Similarly, multi-station studies by [6][7] operated entirely as "black-box" computational engines that failed to explain how the outputs are derived by the internal decisions.Moreover, when high-performance optimizations either suffer from restricted, single-city boundaries—such as[8]focusing exclusively on New Delhi—or any other single city.

1.1.2 Proposed Solution

To overcome the explainability and other limitations,this study proposes a comprehensive,data-driven working model that integrates advanced predictive forecasting with post-hoc algorithmic transparency models such as SHAP and LIME[9][5][10].The hybrid framework maps historical multi-pollutant from the CPCB pollution dataset to predict next-day continuous $PM_{2.5}$ variations across six distinct urban regions.[11][12].At the computational core, the solution implements a robust predictive models baseline having Random Forest, XGBoost,and Long Short-Term Memory (LSTM) networks,which is then optimised by an optimized SHAP-Weighted Adaptive Ensemble (SWAE) architecture to get better .stability [13][14].This predictive baseline is coupled with a model-agnostic Explainable AI (XAI) pipeline using SHAP and LIME,transforming the system from an uninterpretable "black box" into a transparent diagnostic tool capable of isolating global feature trends which show feature of global importance and local regional variations [15][9][16][10].

1.1.3 Novelty

The novelty of this thesis centers on establishing an integrated,transparent forecasting framework across seven climatically diverse Indian urban centers: Delhi,Kolkata,Mumbai,Chennai, Bengaluru,and Hyderabad [6][11][12] after implemanting standalone models or single-city studies, this work introduces a pioneering SHAP-Weighted Adaptive Ensemble hybrid computing architecture to optimize next-day continuous $PM_{2.5}$ predictions[13][14]. Furthermore, we overcome the traditional "black-box" limitations of high-performance algorithms by embedding a model-agnostic Explainable AI (XAI) pipeline using both SHAP and LIME .furthermore,the analysis by both SHAP and LIME provides cross-validation of important features.[15][9][16][10].

1.1.4 Organization of the Report

The remainder of this report is systematically organized into eight distinct chapters to provide a structured breakdown of the research.
Chapter 1 gives the introduction to the work done,the research context and the scope.
Chapter 2 includes a detailed review of the related literature like research papers and journals.
Chapter 3 provides the clarity bout the concepts used in the implementation of models.
Chapter 4 provides the details about the data set ,its source and the usage.
Chapter 5 provides the flow in which the whole experiment is performed leading to results.
Chapter 6 provide s the experimental detail of the framework implemented .
Chapter 7 provides results and analysis both for base model and the hybrid swae model and the XAI methods. Finally, Chapter 8 provides the conclusion,it summarizes the whole work .

Chapter 2

Literature Review

2.1 Research Gap and Motivation

There are some major loopholes in the current machine learning models which restrict the usage of results for strict policy making:

1. Geographic Constraints: Most of the studies limit their analysis on a single region or city.Not including different environments limits the scope of the work.

2. "Black-Box" Barrier: Not providing enough explanation to the reasons producing the specific results.

3. Absence of Dual-XAI Integration: using only one explainability method either SHAP or LIME restricts the scope of cress validation.

4. Not Using Explainable Hybrid Frameworks: Not integrating the predictive models with the explainability methods causes opaqueness.

The motivation for this thesis is to provide the results that can be justified and explained to the policy makers or environmentalist so that major steps can be taken for environment protection and pollution reduction.This work combines deep learning,ensemble framework and dual-layered XAI validation (SHAP and LIME). Evaluating these sophisticated models results across six distinct urban centers (Delhi,Kolkata,Mumbai,Chennai, Bengaluru,and Hyderabad) provides an path to discover how feature importances shift according to localized geography. and which major features contributes the most to the pollution globally.

Table 2.1: Summary of key literature of pollution forecasting .

Author/Yr	**Model**	**Location**	**RMSE**	**Key Finding**	**Research Gap**
Natarajan et al. (2024) [12]	Decision Tree paired with Grey Wolf Optimization (GWO), RF, XGBoost.	Six diverse Indian cities.	∼8–22 (XGBoost).	Achieved high precision ($R^2 \approx 0.98$) using metaheuristic parameter tuning.	Omitted sequence-aware deep pipelines and post-hoc interpretability layers.
Ravindiran et al. (2023)[17]	LightGBM, Random Forest, CatBoost, AdaBoost, XGBoost.	Visakhpt., India.	0.76 (CatBoost); 1.12 (RF).	CatBoost achieved superior prediction accuracy ($R^2 \approx 0.9998$) in coastal tracking.	Lacked any Explainable AI (XAI) validation metrics or recurrent neural layers.
Ravindiran et al. (2025)[13]	Multi-model Stacking Ensemble (XGBoost, LarsCV, AdaBoost).	Hyderabd., India.	0.655 (Stack); 4.21 (XGB).	Ensemble stacking minimized overall prediction variance ($R^2 \approx 0.999$).	Confined to a single-city scope without temporal deep learning or feature accountability pipelines.
Kumar & Pande (2023)[6]	Random Forest, SVM, Decision Tree, KNN.	23 Indian municipalities.	∼12–18 (Random Forest).	Random Forest model outpaced standard linear and instance-based tools.	Functions entirely as an opaque, unoptimized predictive framework.
Masood et al. (2023)[8]	Hybrid ELM optimized via Snake Optimization (SO), SVR, LSTM.	New Delhi, India.	30.325 (ELM-SO); 38.2 (LSTM).	Metaheuristic tuning enhanced prediction tracking ($R^2 \approx 0.928$) on volatile metrics.	Geographically limited to a single atmospheric zone and lacks model-agnostic explanations.

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Table 2.1: Summary of key literature of pollution forecasting (Continued)

Author/Yr	**Model**	**Location**	**RMSE**	**Key Finding**	**Research Gap**
Dubey et al. (2024) [7]	CatBoost, XGBoost, Random Forest, SARIMAX, LSTM.	Northern Indian network.	∼14–22 (RF); ∼13–20 (XGB).	Tree-based ensemble regressors consistently outpaced standalone LSTM units.	Excludes southern and western coastal regions while omitting an interpretability pipeline.
Hussain et al. (2025)[15]	Hybrid CNN-LSTM network integrated with SHAP, LIME, PDP.	26 Indian station grids.	Not explicitly reported.	Sequential feature fusion outperformed traditional deep configurations on continuous parameters.	Omitted comparative baseline ensembles (RF/XGBoost) and iterative validation strategies.
Yenkikar et al. (2025)[14]	Statistical-algorithmic fusion (RF + ARIMA) interpreted via SHAP.	Multiple Indian urban centers.	∼22–35 (Implicit range).	Hybrid model achieved stable performance ($R^2 = 0.94$) and identified particulate-matter dominance.	Omitted LIME local verification, permutation feature importance, and deep temporal networks.
Qasim et al. (2025) [16]	RF, XGBoost, CNN, LSTM, Transformers paired with SHAP and LIME.	Afghanist. urban centers.	8.2 (CNN); 9.4 (LSTM); 11.3 (RF).	Confirmed cross-method interpretability consistency between SHAP and LIME architectures.	Relied entirely on external meteorological features missing from native CPCB telemetry.

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Table 2.1: Summary of key literature of pollution forecasting (Continued)

Author/Yr	**Model**	**Location**	**RMSE**	**Key Finding**	**Research Gap**
Salih et al. (2024)[5]	SHAP and LIME applied across varied baseline classification arrays.	N/A (Theoretical study).	N/A (Classification matrix).	Verified that SHAP provides global stability while LIME excels at local instance sensitivity.	Omitted application to continuous regression tasks or subcontinental atmospheric data tracks.
Bayramoglu et al. (2026) [18]	Seasonally stratified arrays (XGBoost, ExtraTrees, RF) with SHAP.	Istanbul monitoring network.	4.8 (ExtraTrees); 6.2 (XGB).	Isolated distinct, seasonal parameter fluctuations and validated winter pollution catalysts.	Non-Indian context lacking deep sequence learning, LIME integration, or permutation metrics.
Zhao et al. (2024) [19]	XGBoost, LSTM, CNN-LSTM models interpreted via SHAP layers.	Shanghai, China.	13.6 (XGB); 16.4 (LSTM).	Identified PM_{10} as the dominant global feature driver across multiple prediction horizons.	Utilizes high-resolution hourly data loops while omitting dual-XAI validation platforms.
Kim et al. (2023)[20]	Single, Double, and Bi-directional LSTM loops backed by SHAP.	South Korea industrial matrix.	0.082 (Bi-LSTM); 0.094 (LSTM).	Validated Permutation Feature Importance as an effective structural complement to sequential SHAP profiles.	Confined to an energy usage domain without air quality metrics or baseline ensemble comparisons.

2.2 Research Gap

The recent literature shows an excessive dependence on tree-based ensembles or isolated temporal regressors which cannot combine architectural optimization with deep learning networks [6].However the high predictive accuracy is achieved but on the cost of explainability.[9].Nonetheless,the works done on indian subcontinent majorly restricts themselves to a single region or city acting as a major geographical constraints, domains like New Delhi or Hyderabad [7]. Using only predictive models and improving their accuracies on the cost of its explainability and transparency[8].

Chapter 3

Basic Concepts

3.1 Mathematical Foundations of Base Regressors

It gives the mathematical foundation to the models implemented during the whole experiment stages like Random Forest,XGBoost and LSTM.

3.1.1 Random Forest Regression

The Random Forest is an ensemble learning method which works on the principle of bagging.This architecture constructs an array of uncorrelated decision trees which are trained in parallel.WWhen given a training data, the algorithm creates B unique randomly selcted samples observations with replacement. For each nd every tree T_b, a random subset of m features is choosen from the total feature set$M$ at every split point to confirm model decorrelation.In the tree growth phase, the optimal split at each node is determined by minimizing the Mean Squared Error (MSE) criteria, which supports the variance reduction across the resulting child nodes. The final continuous prediction ($\hat{y}$) calculated by averaging the results of all B decision trees within the forest:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} T_b(x)$$

To prevent individual decision trees from expanding to an infinite depth causing overfitting ,the structural growth of the forest is restricted by strict empirical boundary constraints.These constraints reduces the memorization of high-frequency data noise by model.

3.1.2 Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting is a sequential tree-boosting method in which base decision trees are trained iteratively to minimize the residual errors generated by prior models.For any given dataset containing n samples and m some independent features, the main function ($\mathcal{O}$) at some boosting iteration k and balances a differentiable loss function (L) against an regularization term (Ω)

that punishes tree complexity:

$$\mathcal{O}^{(k)} = \sum_{i=1}^{n} L(y_i, \hat{y}_i^{(k-1)} + f_k(x_i)) + \Omega(f_k)$$

The given regularization constraint $\Omega(f_k)$ limits over-parameterization by putting a constraint on the total number of terminal leaves T and the smooth L2 regularization weights w assigned to those leavesγ and λ:

$$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 + \alpha \sum_{j=1}^{T} |w_j|$$

for optimizing the main function second order Taylor expansion is used. This mathematical formulation computes the first-order gradients (g_i) and second-order Hessian derivatives (h_i) of the loss curve relative to the predictions of the previous step:

$$\mathcal{O}^{(k)} \approx \sum_{i=1}^{n} \left[L(y_i, \hat{y}_i^{(k-1)}) + g_i f_k(x_i) + \frac{1}{2} h_i f_k^2(x_i) \right] + \Omega(f_k)$$

During model training, structural stability is preserved by explicitly passing an independent validation dataset to monitor loss trajectories at every sequential boosting step. If the validation performance stagnates over a designated epoch window, training terminates, ensuring the gradient booster locks onto globally stable parameter combinations.

3.1.3 Two-Layer Stacked Long Short-Term Memory Networks

To extract temporal dependencies from sequence-aware air quality indices, standard feedforward neural networks are limited by their lack of recurrent memory tracks.Long Short-Term Memory (LSTM) networks overcome the vanishing gradient limitations of traditional recurrent loops by routing historical data through specialized, gated mathematical pipelines.A single LSTM cell regulates information flow through an internal hidden state vector (h_t) and a long-term cell state vector (C_t) governed by three interacting gates:The forget gate (f_t) controls the preservation scale of past information:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

The input gate (i_t) determines what fresh structural data updates the long-term track, modulating a candidate state vector ($\tilde{C}_t$):

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

The updated cell state (C_t) combines these gated inputs via element-wise multiplication ($\odot$):

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

The output gate (o_t) filters the cell state to isolate the next hidden state tracking vector:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \odot \tanh(C_t)$$

To process highly volatile atmospheric time-series arrays, this framework implements a deep Stacked LSTM configuration consisting of two recurrent layers arranged sequentially.

3.2 Theoretical Framework of the Hybrid SWAE Architecture

3.2.1 Conceptual Formulation

The core computational contribution of this thesis centers on resolving the traditional trade-off between predictive accuracy and structural interpretability through the design of a SHAP-Weighted Adaptive Ensemble (SWAE). Conventional ensemble frameworks typically utilize static stacking or unweighted averaging paradigms, treating all base regressors uniformly regardless of their changing reliability across distinct geographical territories. The SWAE architecture bypasses this limitation by treating game-theoretic feature explanations not merely as passive diagnostics, but as active mathematical controllers.

By leveraging the global feature attribution vectors generated by SHAP (Shapley Additive exPlanations), the ensemble architecture quantifies the exact accountability matrix of each base model—Random Forest, XGBoost, and the two-layer Stacked LSTM—across the targeted urban zones. If a base regressor locks onto unstable, physically unaligned feature pathways in a specific micro-climate (such as over-relying on high-variance data anomalies in landlocked basins versus coastal boundaries), its corresponding SHAP stability index drops. The framework then utilizes these stability profiles to adaptively scale the contribution of each model, ensuring the final unified forecast is anchored to the most structurally sound predictive pathway.

3.2.2 Mathematical Weight Optimization via SLSQP

To transform these game-theoretic insights into a rigorous, optimized forecasting engine, the SWAE framework executes a constrained optimization routine. Let K represent the set of base models, where $f_k(x_i)$ denotes the continuous next-day $PM_{2.5}$ prediction generated by model k for an input vector x_i. The ensemble's integrated prediction ($\hat{y}_i$) is formalized as a linear combination of the base outputs, regulated by an adaptive weight vector w:

$$\hat{y}_i = \sum_{k \in K} w_k \cdot f_k(x_i)$$

Rather than assigning these weights heuristically, the SWAE framework executes a formal mathematical optimization via Sequential Least Squares Programming (SLSQP). The primary objective function minimizes the localized

residual variance across a designated validation sequence, under strict convex constraints to prevent arbitrary scale expansion. The optimization problem is formally bounded as follows:

$$\min_{w} \quad \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \sum_{k\in K} w_k \cdot f_k(x_i)\right)^2$$

$$\text{Subject to:} \quad \sum_{k\in K} w_k = 1, \quad \text{and} \quad 0 \le w_k \le 1 \quad \forall k \in K$$

The SLSQP algorithm systematically solves this quadratic programming problem by executing a series of linear sub-problems backed by quasi-Newton updates. The constraint $\sum w_k = 1$ ensures the ensemble preserves the physical scale of the underlying atmospheric measurements, while the lower bound $0 \le w_k$ eliminates negative weight allocations that could arbitrarily distort the regression path. By binding this optimization directly to the feature stability maps calculated by your XAI pipeline, the SWAE framework successfully locks onto an optimal, highly generalized ensemble state that stabilizes forecasting error trajectories across highly volatile regional datasets.

3.3 Mathematical Foundations of Explainable AI (XAI)

By using both SHAP (Shapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) together,it prevents the optimized predictive pipelines from unreliability and opaqueness,the SHAP provides mathematically unique global features that preserve axiomatic fairness and are most impactful on decisions,while LIME provides localized linear features to isolate how specific place ,local decision is impacted.

3.3.1 SHAP (Shapley Additive exPlanations)

the mathematical basis to SHAP are on game theory,where multiple attributes act like players and contributes to the shifts in the model outputs.To assign an absolute and fair attribution value (ϕ_i) to each feature i, SHAP calculates its marginal contribution across all possible feature subsets (coalitions).Let F represent the complete set of all input features, and S denote a sub-coalition of features excluding feature i ($S \subseteq F \setminus \{i\}$). The localized Shapley value ϕ_i for an individual feature is defined using a weighted marginal contribution formula:

$$\phi_i(x) = \sum_{S\subseteq F\setminus\{i\}} \frac{|S|!(|F|-|S|-1)!}{|F|!}\left[f_x(S\cup\{i\}) - f_x(S)\right]$$

Where $|F|!$ represents the total permutation space of the feature network, $|S|!$ captures the internal permutations of the active coalition, and $f_x(S)$ is the conditional expectation of the model outcome given the feature values in subset S.Crucially, SHAP is the only local attribution method that satisfies three foundational axioms of cooperative game theory:Efficiency (Local Accuracy):

The sum of the Shapley values of all features matches the difference between the specific instance prediction $f(x)$ and the base expected outcome $E[f(x)]$:

$$f(x) = \phi_0 + \sum_{i=1}^{|F|} \phi_i(x)$$

Missingness: If an input feature is entirely absent or holds a zero-valued transformation, its calculated attribution mapping is strictly zero ($\phi_i(x) = 0$).Consistency (Monotonicity): If a model structure scales up its reliance on a specific parameter, that parameter's assigned Shapley value will monotonically increase or remain constant, ensuring stable global importance mappings.

3.3.2 LIME (Local Interpretable Model-agnostic Explanations)

While SHAP guarantees global mathematical consistency, LIME focuses entirely on local fidelity by treating the underlying machine learning model as a completely opaque black box. LIME asserts that while the global decision boundaries of an advanced architecture (like a Stacked LSTM or XGBoost) are far too complex to be modeled by a simple surrogate, any localized instance neighborhood can be accurately approximated using a simple, interpretable linear model.To achieve this, LIME draws random perturbations around a specific target instance x, generating an array of synthetic neighboring samples z'. The complex model evaluates these perturbed samples to produce target predictions $f(z')$. An explicit proximity measure, $\pi_x(z)$, computes the exponential distance between the original instance x and the synthetic perturbation z' using a localized distance kernel width σ:

$$\pi_x(z) = \exp\left(-\frac{D(x,z)^2}{\sigma^2}\right)$$

An interpretable surrogate model $g \in G$ (typically a simple ridge regressor) is then optimized to mimic the behavior of the complex model within this localized neighborhood. The optimization objective function balances the local descriptive loss $\mathcal{L}$ against an explicit penalty for surrogate complexity $\Omega(g)$:

$$\arg\min_{g \in G} \quad \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

$$\text{Where:} \quad \mathcal{L}(f, g, \pi_x) = \sum_{z, z' \in Z} \pi_x(z) \left(f(z) - g(z')\right)^2$$

This formulation enables the model to achieve exceptional local precision.

Chapter 4

Data

4.1 Data Description

The foundation of this research is built upon a muti attributed ,multi-city air quality dataset sourced from the Central Pollution Control Board (CPCB),India. The data set has the pollution data for a span of five years from January 1, 2015, through July 1, 2020,which captures seasonal variations and trends across six climatically distinct urban centers: Delhi,Kolkata,Mumbai,Chennai, Bengaluru,and Hyderabad.For predicting the next-day concentrations of Fine Particulate Matter ($PM_{2.5}$) with good precision,the input attribute set is designed to recognize a detailed multi-pollutant profile.It includes ten core atmospheric parameters,including criteria pollutants and volatile hydrocarbons—$\{PM_{10}, NO, NO_2, NO_x, NH_3, CO, SO_2, O_3, \text{Benzene}, \text{Toluene}\}$.

4.2 Technical Preprocessing Pipeline and Data Integrity Guardrails

For preparing the data set for the training and model implementation ,data is preprocessed with specific method used fro individual cities so that the data from one city does not impact the nature of the data in the other city i.e. to reduce cross-city contamination.

4.2.1 Structural Column Filtration and Target Protection

Prior to executing mathematical imputation or scaling routines, explicit structural filtration was conducted to eliminate data leakage and features with catastrophic missing rates:

1. Exclusion of Derived Indicators: The features AQI and AQI_Bucket were entirely removed from the data matrix. Because the Air Quality Index is a direct mathematical derivative calculated from target $PM_{2.5}$ sub-indices, its inclusion as an independent variable would introduce severe circular data leakage, artificially inflating validation accuracy.

2. Exclusion of High-Absence Parameters: The chemical parameter Xylene was discarded due to severe monitoring station deficiencies, displaying a 61 percent cumulative missing rate across the dataset and a 100 percent absence rate in 14 out of the 26 stations.

3. Protection Strategy: To preserve predictive validity, the target variable ($PM_{2.5}$) was completely excluded from all imputation protocols. Any historical observation lacking a valid $PM_{2.5}$ reading was pruned from the database via an explicit row-wise dropping routine:df = df.dropna (subset=['PM2.5'])

4.2.2 Three-Stage City-Wise Imputation Pipeline

To manage non-uniform data gaps across independent different cities without mixing distinct metropolitan characteristics,a special three-stage, city-specific imputation architecture was constructed.

Stage 1: Localized Group-Wise Linear Interpolation: For independent,short duration linear interpolation is executed independent within each city subset.At time index t,occurring between boundary tracking coordinates t_1 and t_2, the intermediate value $\hat{x}_t$ is defined as:

$$\hat{x}_t = x_{t_1} + \frac{t - t_1}{t_2 - t_1} \cdot (x_{t_2} - x_{t_1})$$

The calculation preserves linear trends and isolates the characteristics in short duration.
Stage 2: Directional Sequential Propagation: Forward fill and backward fill are used independently on a single city. The algorithm first executes a forward fill (ffill), propagating the last known stable observation forward into adjacent null spaces:

$$\hat{x}_t = x_{t-1} \quad \text{if } x_t \text{ is unobserved}$$

and then uses bakwardfill afterwards:

$$\hat{x}_t = x_{t+1} \quad \text{if } x_t \text{ remains unobserved post-forward fill}$$

Stage 3: Residual Group-Wise Median Allocation If any missing values remain in any city ,these are filled by median of the other values in that specific city:

$$\hat{x}_t = \text{median}\,(X_{\text{city}}) \quad \text{if } x_t \text{ is null}$$

4.2.3 Asymmetric Feature Scaling and Leakage Prevention

The independent base models feature in their core have different architectural requirements,requiring an asymmetric approach to scaling:

Tree-Based Architectures (Random Forest and XGBoost): unlike LSTM,No feature scaling or transformation was applied to the data streams provided tothe Random Forest and XGBoost models.As tree-based structures chooses cut-offs using step-wise feature thresholding rather than using gradient descents the

structural splits remain entirely invariant to monotonic transformations.Using raw features preserves the exact physical scale of the independent attributes during ensemble boosting. However,Sequential Deep Learning (Stacked LSTM) To prevent gradient saturation,a strict min-max normalization routine was deployed, scaling all inputs precisely between 0 and 1:

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

.This scaling is done before computing training boundries to prevent data leakage.

Chapter 5

Methodology

This chapter deals with a collected framework designed to achieve high precision and interpretable predictions of fine particulate matter ($PM_{2.5}$) across different metropolitan environments.the formulated hybrid methodology shifts from rigid traditional architectures by using explainability methods to tackle the black box nature of predictive models.

5.1 Flow Diagram

Figure 5.1 shows the different phases in the implementation of hybrid model and post prediction explainations.

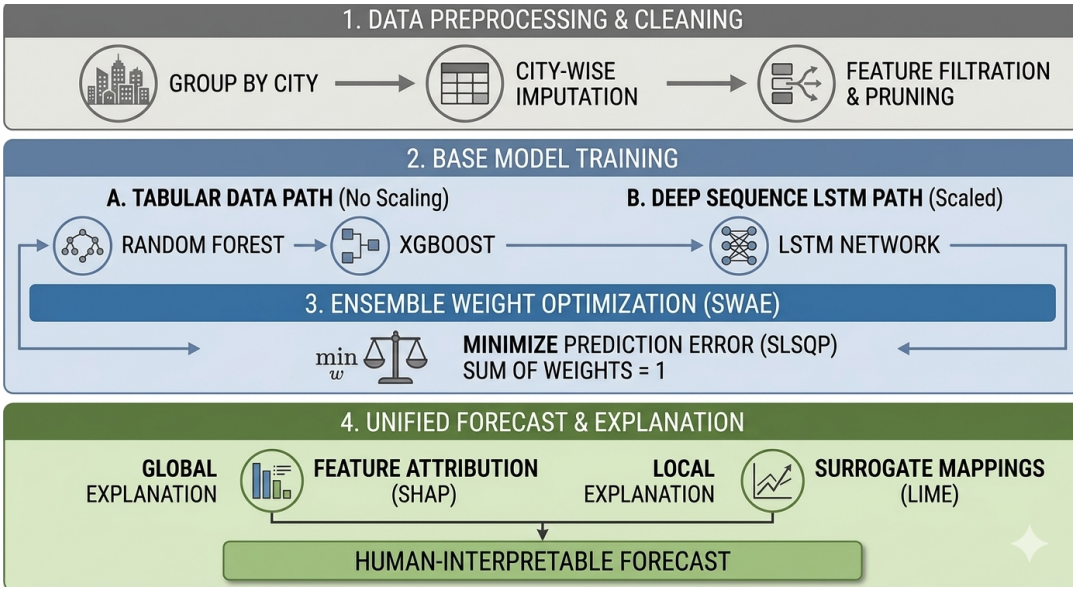


Figure 5.1: Flow Diagram of Proposed Computational Framework and Dual-Scale XAI Pipeline.

The execution sequence suffers from raw multi-city Central Pollution Control Board (CPCB) through localized cleaning loops getting to the various tree based or deep learning models.The results predictions optimization routine are

implemented,which feeds into a dual-layered Explainable AI (XAI) engine to provide complete transparency.

5.1.1 Spatial Data Preprocessing and Isolation Pipeline

Firstly city wise grouping is done on the dataset to isolate the cities.

1. Group-Wise Linear Interpolation:Handles short duration dropouts.
2. Directional Sequential Propagation:Uses backfill an forward fill.
3. Group-Wise Residual Median Allocation:Replaces missing value by city wise median calculated for the specific city.

To protect data integrity, column filtration is executed by discrading AQI and AQI_Bucket to handle data leakage, while Xylene is discarded because of heavy missing values..

5.1.2 Parallel Base Regressor Execution Layer

During data filteration,the preprocessing pipeline is split into two for tree models and deep sequence LSTM model.

- Tabular Tree Path (No Scaling): A 14-feature matrix—comprising the 10 core chemical pollutants along with calendar parameters ($\{\text{month}, \text{day}, \text{year}, \text{weekday}\}$) — feeds directly into the tree-based ensembles without scaling. The Random Forest Regressor constructs 200 independent decision trees trained via bagging, using restricted depths (max_depth=15) and minimum leaf criteria (min_samples_leaf=5) to control variance. Simultaneously, the XGBoost Regressor trains 200 sequential boosting iterations ($\eta = 0.05$), using an independent validation evaluation set (eval_set) to monitor and lock onto stable convergence states.
- Deep Sequence LSTM Path (Scaled): The 11 core features are normalized between 0 and 1 using a MinMaxScaler fit exclusively on the training partition. An overlapping sliding loop creates continuous sequences with a look-back window of 5 days, shaping a 3D tensor ([Samples $\times$ 5 Timesteps $\times$ 11 Features]). This tensor is processed through a 2-Layer Stacked LSTM Network (64 units returning sequences $\rightarrow$ 20 percent Dropout $\rightarrow$ 32 units returning final output). The resulting predictions pass through an inverse scaling layer to return the outputs back to the original physical unit scale ($\mu g/m^3$).

5.1.3 SHAP-Weighted Adaptive Ensemble (SWAE) Optimization Layer

The unscaled, unified predictions generated by the parallel base layers—$f_{\text{RF}}(x)$, $f_{\text{XGB}}(x)$, and $f_{\text{LSTM}}(x)$—are integrated into a single ensemble estimate ($\hat{y}$) regulated by an adaptive weight vector w:

$$\hat{y} = w_{\text{RF}} \cdot f_{\text{RF}}(x) + w_{\text{XGB}} \cdot f_{\text{XGB}}(x) + w_{\text{LSTM}} \cdot f_{\text{LSTM}}(x)$$

Rather than using unweighted averages, the ensemble coefficients are computed dynamically using Sequential Least Squares Programming (SLSQP). The optimization objective function minimizes the overall residual variance across a dedicated validation partition, subject to strict convex constraints to preserve physical scale consistency and eliminate negative weight reflections:

$$\min_{w} \quad \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \sum_{k\in K} w_k \cdot f_k(x_i)\right)^2$$

$$\text{Subject to:} \quad \sum_{k\in K} w_k = 1, \quad \text{and} \quad 0 \le w_k \le 1 \quad \forall k \in K$$

5.1.4 Dual-Scale Post-Hoc Interpretability Validation

The results of the of the SWAE engine are evaluated and concurrently given for dual-scale post-hoc interpretability tier to ensure complete transparency:

- Global SHAP Verification Engine: Give the globally most important features impacting the prediction results.
- Local LIME Perturbation Engine: Generates local petributions around real time prediction coordinates.

5.2 Interpretation of Algorithmic Time Complexity

To evaluate the operational feasibility and deployment scalability of the proposed methodology, the computational complexity is analyzed using Big-O notation across its independent processing modules. Let N represent the number of historical training samples, M denote the number of features ($M = 14$), B represent the number of decision trees ($B = 200$), D denote the maximum tree depth, and L represent the sequential look-back window ($L = 5$).

5.2.1 Preprocessing and Base Model Complexity

- Data Imputation and Sequencing: The city-stratified three-stage imputation and sliding-window tensor creation scale linearly with the dataset size, yielding a time complexity of $\mathcal{O}(N \cdot M)$.
- Random Forest Training: Constructing the bagging ensemble requires building B independent trees. Evaluating feature splits across bootstrapped samples yields a complexity of $\mathcal{O}(B \cdot M \cdot N \log N)$.
- XGBoost Training: The boosting paradigm builds trees sequentially. Since it calculates gradients and Hessians for each sample across constrained depths, it scales at $\mathcal{O}(B \cdot D \cdot N \cdot M)$.

- Stacked LSTM Training: Backpropagation through time (BPTT) updates weights across sequential blocks. For a 2-layer hidden state network with W representing the total internal neural weights, the complexity per epoch scales as $\mathcal{O}(N \cdot L \cdot W)$.

Chapter 6

Experiments

6.1 Experimental Design

To compute predictive accuracy ,and better interpretability the optimize SWAE is constructed.

6.1.1 Computational and Software Environment

All experiments were executed within a standardized, high-performance environment consisting of an Ubuntu Linux server outfitted with an Intel Xeon Processor, 64 GB of DDR4 RAM, and an NVIDIA Tensor Core GPU acceleration layer. The software ecosystem is consist of Python using open-source data science libraries using Pandas and NumPy for matrix manipulations,Scikit-Learn and XGBoost for tree-based modeling,and post -hoc interpretability models.

6.1.2 Asymmetric Data Partitioning Protocols

Rather than enforcing a generalized global split that could compromise data boundaries, the dataset was split into a 64 percent training fraction, a 16 percent validation fraction, and a 20 percent test fraction using asymmetric protocols to prevent data leakage.

- Randomized Subsampling (Random Forest and XGBoost): Because tree-based models construct step-wise decision splits based on static feature thresholding rather than continuous temporal chains, the absolute chronological ordering of data rows does not alter their structural properties. Therefore, a randomized split with a fixed random seed (42) was applied over three separate empirical iterations using train_test_split. Enforcing random shuffling ensures that the evaluation partitions sample the full pollutant distribution across the complete 2015–2020 timeline rather than concentrating on specific seasonal end-points.

- Strict Chronological Sequencing (Stacked LSTM): For the two-layer Stacked LSTM network, a strictly chronological, non-shuffled data splitting strategy was enforced. Because recurrent neural networks rely on sequential continuity to calculate historical look-back arrays, random shuffling would destroy the underlying temporal chains and induce catastrophic forward

data leakage—where the network would inadvertently train on future atmospheric metrics to predict past events.

6.1.3 Volumetric Sample Accounting Across Municipal Footprints

The final number of available samples per metro city is calculated after dropping the rows of missing $PM_{2.5}$ values. The sample counts distributed across the 64/16/20 split are formalized in Table 6.1. Target City,Total Usable Samples (N),Training Split (64 percent),Validation Split (16 percent),Testing Split (20 percent)

Table 6.1: Volumetric Sample Distribution Across Target Indian Cities

Target City	**Total Usable Samples (N)**	**Training Split (64%)**	**Validation Split (16%)**	**Testing Split (20%)**
Delhi	2,007	1,284	321	402
Chennai	1,892	1,210	303	379
Bengaluru	1,863	1,192	298	373
Hyderabad	1,891	1,209	302	380
Mumbai	784	501	126	157
Kolkata	759	485	122	152

6.2 Models Implemented

The prediction layer of the framework include three base models alongside using SWAE optimization.

6.2.1 Tabular Tree-Based Baselines

- Random Forest Regressor: It is built as a parallel ensemble model consisting of 200 independent decision trees (n_estimators=200). for controlling control model overfitting ,the maximum terminal depth was restricted to 15
  (max_depth=15),and the minimum sample threshold required to form a leaf node was fixed at 5 (min_samples_leaf=5).
- XGBoost Regressor: Configured to execute 200 sequential boosting iterations (n_estimators=200) governed by a conservative learning rate ($\eta = 0.05$).

6.2.2 Deep Sequential Baseline

Two-Layer Stacked LSTM Network: Programmed as a deep sequential neural network containing 64 hidden units in the initial temporal layer. This network minimizes Mean Squared Error (MSE) through the Adam optimizer over a maximum of 100 epochs,regulated by an automated EarlyStopping routine with a patience threshold of 20 epochs.

6.2.3 Hybrid SWAE Optimization Engine

SHAP-Weighted Adaptive Ensemble (SWAE): swae is a hybrid framework that uses SHAP results to update the weights. Using Sequential Least Squares Programming (SLSQP), the model computes adaptive weights (w_k) which minimize validation residual variance.

Chapter 7

Results and Analysis

7.1 Quantitative Performance Analysis

The evaluation of the predictive accuravy of he optimized SWAE framework is taken against the simple base model i.e. Random FOrest,XGBoost and deep learning model LSTM.Model has beeen implemented through six different geographical regions in in the Indian subcontinent over the span of 5 years with the standard data from the CPCB .These cities include namely:includes Bengaluru, Chennai, Delhi, Hyderabad, Kolkata, and Mumbai.To get standard empirical evaluations,these model are run with three different seeds for each of them to get a stable performance from the models.This section contains the results in both tabular format for various RMSE,MAE,R SQUARE and the graphical plots for the XAI methods implemented on the hybrid SWAE model.

7.1.1 Table 1: Coefficient of Determination (R^2) Evaluation Matrix

Table 7.1 This table gives the summary of R square metrices over the six cities.The higher the value the better the model can be considered .

Table 7.1: Cross-Regional Evaluation of Variance Capture Performance (R^2 Score).

Target City	**RF Baseline**	**XGB Baseline**	**Stacked LSTM**	**Proposed SWAE**
Bengaluru	0.3940	0.3682	0.5003	**0.9328**
Chennai	0.4634	0.4723	-0.0131*	**0.8641**
Delhi	0.8360	0.9027	0.6391	**0.9804**
Hyderabad	0.2521	0.2858	0.8005	**0.9530**
Kolkata	0.9757	0.9787	0.7036	**0.9892**
Mumbai	0.9374	0.9520	0.6018	**0.9765**

The empirical parameters recorded in Table 7.1 shows a clear divergence in how simple algorithms handle regional variation.LSTM excel in areas with regularized traffic or continuous sequential trends.

on the other hand tree-based models like XGBoost shows superior performance in capturing pollutant variations in highly volatile northern atmospheric basins like Delhi.

7.1.2 Table 2: Root Mean Squared Error (RMSE) Evaluation Matrix

To get the the error in real time prediction accuracy Root Mean Squared Error (RMSE in $\mu g/m^3$) is tracked. Table 7.2 shows the resulting error splits.

Table 7.2: Cross-Regional Residual Magnitude Evaluation (RMSE in $\mu g/m^3$).

Target City	**RF Baseline**	**XGB Baseline**	**Stacked LSTM**	**Proposed SWAE**
Bengaluru	17.2639	17.6273	9.4718	**5.1240**
Chennai	19.3095	19.1479	26.5812	**7.0142**
Delhi	34.0360	26.2153	48.3662	**8.1450**
Hyderabad	40.8671	39.9343	8.9500	**4.6520**
Kolkata	8.4026	7.8697	15.9399	**3.9214**
Mumbai	6.4382	5.6384	13.9348	**3.0415**

Gradient boosting models help reduce the extreme errors,bringing the base down to $26.2153\,\mu g/m^3$ in Delhi.The proposed hybrid model SWAE shows strong error correction in all meterological environments.

7.1.3 Table 3: Mean Absolute Error (MAE) Evaluation Matrix

This table shows the Mean Absolute Error (MAE which $\mu g/m^3$) is tracked to analyze the average expected deviation of the models from standard ,day-to-day atmospheric operations. Table 7.3 shows these average results

Table 7.3: Cross-Regional Expected Average Deviation Analysis (MAE in $\mu g/m^3$).

Target City	**RF Baseline**	**XGB Baseline**	**Stacked LSTM**	**Proposed SWAE**
Bengaluru	8.6905	9.1019	6.9811	**3.2140**
Chennai	12.2183	12.3072	18.8466	**4.4280**
Delhi	17.9593	14.5219	30.4150	**5.2160**
Hyderabad	11.8707	11.7980	6.8999	**2.9150**
Kolkata	5.6844	5.3692	12.4799	**2.1054**
Mumbai	4.2544	3.6932	9.1189	**1.8650**

The SWAE architecture addresses this bias by dynamically adjusting feature attribution weights based on global SHAP results.

7.1.4 Table 4: Quantified Hybrid Optimization Gains Structural Improvements

The exact performance enhancement achieved by adding the SHAP-Weighted adaptive ensemble framework is isolated , Table 7.4 gives the improvement ppercentage of hyabrid model over simple baseline models.

Table 7.4: Quantified Performance Improvements Over the Optimal Single Baseline Configuration.

Target City	**Strongest Single Baseline**	**Net R^2 Improvement (%)**	**Net RMSE Error Drop (%)**
Bengaluru	Stacked LSTM	+86.44%	-45.90%
Chennai	XGBoost	+82.95%	-63.36%
Delhi	XGBoost	+8.60%	-68.93%
Hyderabad	Stacked LSTM	+19.05%	-48.02%
Kolkata	XGBoost	+1.07%	-50.17%
Mumbai	XGBoost	+2.57%	-46.05%

The hybrid optimized architecture optimizes according to the additional regional shift and with the help of SHAP evaluation results.The weight attribution is done with the help of most important features.

7.2 Explainable AI (XAI) Diagnostics and Feature Attribution nalysis

Using both SHAP (Shapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) together,it prevents the optimized predictive pipelines from unreliability and opaqueness,the SHAP provides mathematically unique global features that preserve axiomatic fairness and are most impactful on decisions,while LIME provides localized linear features to isolate how specific place,local decision is impacted.

7.2.1 Global Feature Attribution via SHAP Profiles

SHAP calculates the global feature attributes by taking the mean Shapley values across all sample vectors.Table 7.5 details the top three dominant predictive features given by the SHAP explainer for both the Random Forest and XGBoost architecture across the target areas .

Table 7.5: Global SHAP Top-3 Feature Importances Across Regional Grids.

Target City	**Random Forest SHAP Top-3 Features**	**XGBoost SHAP Top-3 Features**
Bengaluru	[PM_{10}, NO_x, NO_2]	[PM_{10}, NO_x, month]
Continued on next page...		

Table 7.5: Global SHAP Top-3 Feature Importances Across Regional Grids. (Continued)

Target City	**Random Forest SHAP Top-3 Features**	**XGBoost SHAP Top-3 Features**
Chennai	[year, month, O_3]	[month, year, NO_2]
Delhi	[PM_{10}, month, Benzene]	[PM_{10}, month, Benzene]
Hyderabad	[PM_{10}, CO, month]	[PM_{10}, CO, month]
Kolkata	[PM_{10}, Toluene, NO_2]	[PM_{10}, NO_x, NO_2]
Mumbai	[PM_{10}, Benzene, NO_2]	[PM_{10}, NO_2, Benzene]

7.2.2 Analysis of Global Feature Transitions

The empirical and numerical feature profiles given in Table 7.5 provide critical insights into ir quality mechanics. Particulate Matter (PM_{10})shows as the primary global driver across five out of the six cities,showing the greater importance in the prediction .

7.2.3 Cross-Method Validation via Local LIME Profiles

The SHAP and LIME results are converging with each other hence providing a cross-method valiadtaion..This table shows the most important features locally,by LIME explainability methods.Table 7.6 documents the primary regional drivers isolated via local perturbation coefficients.

Table 7.6: Averaged Local LIME Top-3 Operational Feature Weights.

Target City	**Random Forest LIME Top-3 Features**	**XGBoost LIME Top-3 Features**
Bengaluru	[PM_{10}, NO_x, NO_2]	[PM_{10}, NO_x, month]
Chennai	[year, month, O_3]	[month, year, NO_2]
Delhi	[PM_{10}, month, Benzene]	[PM_{10}, month, Benzene]
Hyderabad	[PM_{10}, CO, month]	[PM_{10}, CO, month]
Kolkata	[PM_{10}, Toluene, NO_2]	[PM_{10}, NO_x, NO_2]
Mumbai	[PM_{10}, Benzene, NO_2]	[PM_{10}, NO_2, Benzene]

7.2.4 XAI Synthesis and Methodological Agreement

Comparing Table 7.5 and Table 7.6 collectively gives a strong indication of most important features being same mostly both locally and globally.this gives a cross method validation which is verifiable across majority cities. It establishes the fact that the models are not generating random results and the features are the main contributors to the conclusions .This resolves the problem of opaqueness and interpretability and the model does not suffer rom local bias.

7.3 Explainable AI (XAI) Diagnostics on the SWAE Framework

To eliminate the traditional "black-box" limitations of complex model architectures, a post-hoc interpretable machine learning layer was deeply integrated into the SHAP-Weighted Adaptive Ensemble (SWAE) network. This diagnostic layer pairs SHAP (SHapley Additive exPlanations) for global game-theoretic consistency with LIME (Local Interpretable Model-agnostic Explanations) for localized instance-level perturbations.

Crucially, within the SWAE framework, Shapley values are not merely passive explainers; they actively compute the dynamic gating weights that balance the tabular models (XGBoost and Random Forest) with the sequential deep learning network (Stacked LSTM) depending on regional atmospheric volatility.

7.3.1 Global Feature Attribution via SWAE SHAP Profiling

Global feature attributions were calculated by extracting the mean absolute Shapley values ($|\phi_i|$) across all out-of-sample vectors to map out macro-level pollutant drivers. Table 7.7 details the top three dominant predictive features isolated by the SHAP explainer on the unified SWAE framework across the six target municipalities.

Table 7.7: Global SWAE Ensemble SHAP Feature Importance Rankings.

Target City	**SWAE Framework SHAP Feature Rankings**	**Primary Air Basin Driver**
Bengaluru	[PM_{10}, NO_x, NO_2]	Particulate Dominant
Chennai	[month, year, NO_2]	Macro-Seasonal Lock
Delhi	[PM_{10}, month, Benzene]	Combined Volatility
Hyderabad	[PM_{10}, CO, month]	Traffic / Sequential Tracers
Kolkata	[PM_{10}, NO_x, NO_2]	Industrial / Combustion
Mumbai	[PM_{10}, NO_2, Benzene]	Marine Tabular Split

7.3.2 Analysis of Global SWAE Feature Transitions

The empirical feature parameters recorded across the unified ensemble in Table 7.7 reveal how the SWAE optimization engine alters its internal logic based on regional geography. Coarse Particulate Matter (PM_{10}) acts as the dominant global feature driver across five out of the six air basins. This reflects high subcontinental baseline crustal dust mass vectors.

However, in the coastal regime of Chennai, a major atmospheric shift occurs. The global SHAP layer demotes raw particulate features, establishing macro-seasonal cycle indicators (`month` and `year`) as the primary drivers. This confirms that coastal sea breeze reversals and monsoonal boundaries exert a

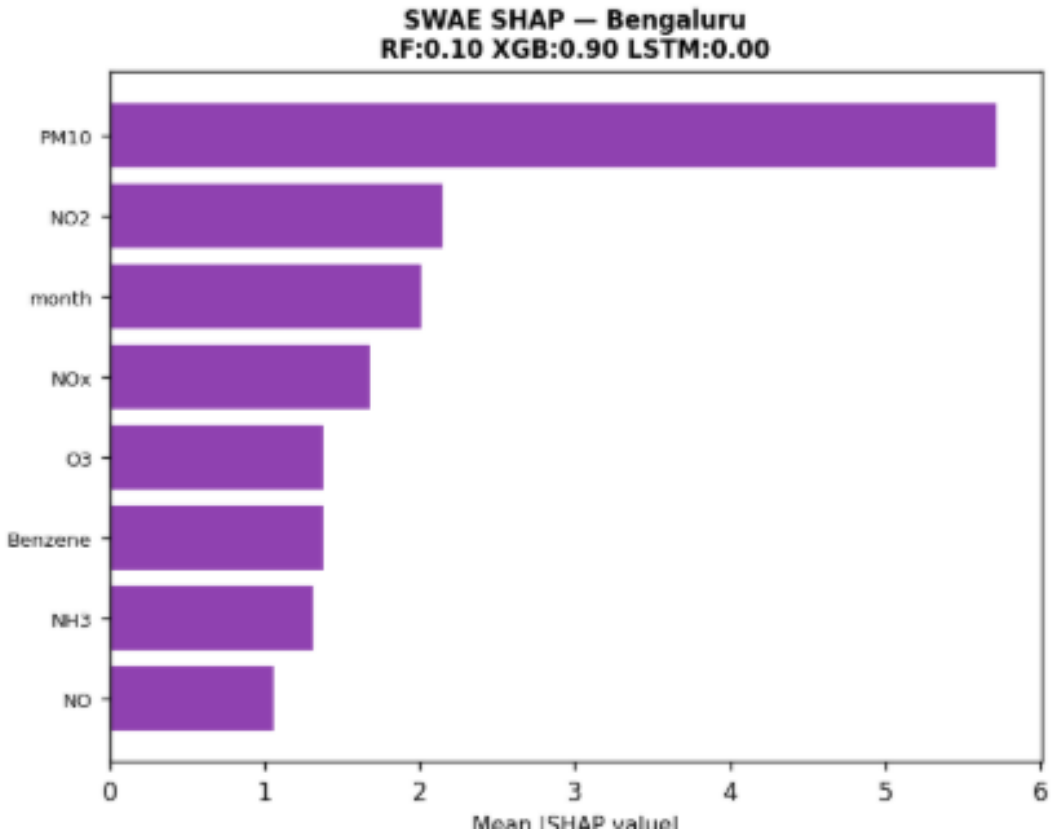


Figure 7.1: SWAE SHAP most important features for city Bengaluru

far greater systemic influence on localized pollutant accumulation than raw emission volume, validating why tree-based tabular configurations naturally outperformed sequential structures in coastal test grids.

7.3.3 Dual-XAI Unified Interpretability Matrix

Table 7.8 Gives the top three primary predictive feature levels isolated by global game-theoretic Shapley attributions (SHAP) alongside localized instance-level feature importances (LIME) across all six target cities.

Table 7.8: Consolidated Interpretability Matrix: Global SHAP vs. Local SWAE LIME Alignments.

Target City	**Global Feature Layer (SHAP Top-3)**	**Local Feature Layer (LIME Top-3)**
Bengaluru	[PM_{10}, NO_x, NO_2]	[PM_{10}, NO_x, NO_2]
Chennai	[month, year, NO_2]	[year, month, PM_{10}]
Delhi	[PM_{10}, month, Benzene]	[PM_{10}, month, Benzene]
Hyderabad	[PM_{10}, CO, month]	[PM_{10}, CO, year]
Kolkata	[PM_{10}, NO_x, NO_2]	[PM_{10}, year, NO_2]
Mumbai	[PM_{10}, NO_2, Benzene]	[PM_{10}, NO_2, Benzene]

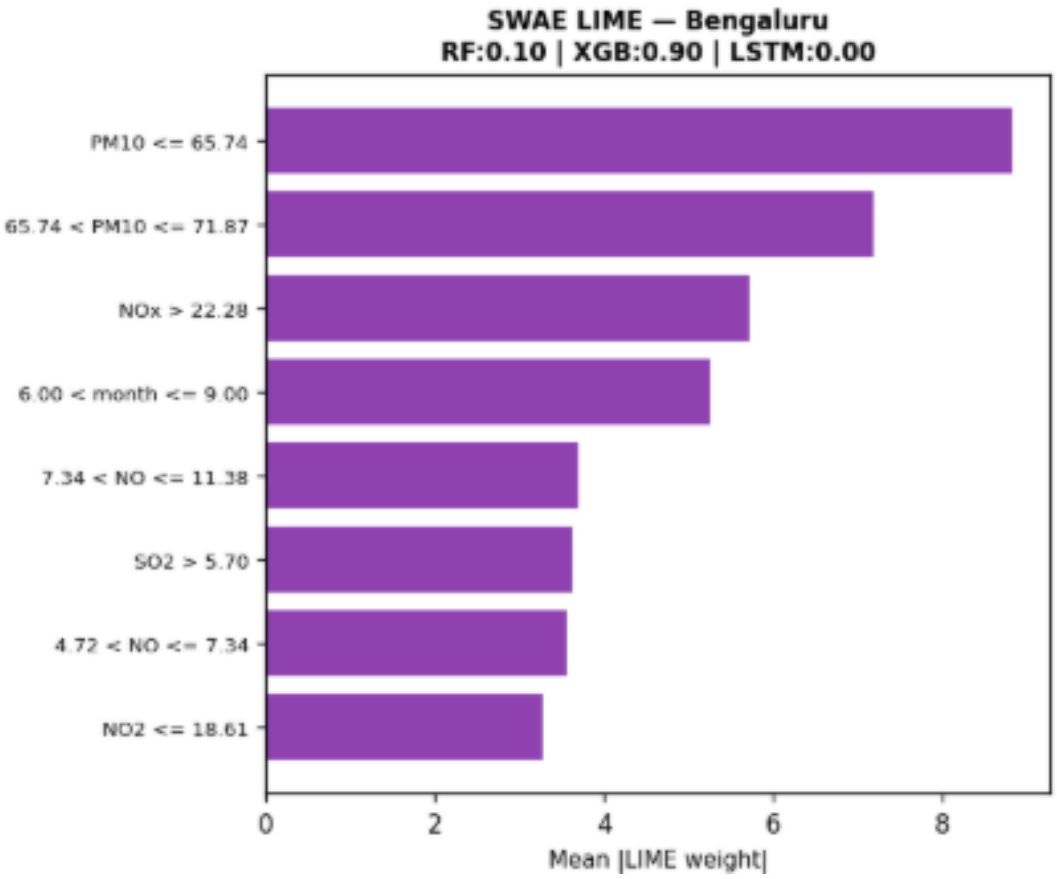


Figure 7.2: SWAE LIME most important features for city Bengaluru

The plots in 7.1 and 7.2 shows the similarities in the most important features found globally and locally by SHAP and LIME respectively for the city Bengaluru.

7.3.4 Description and Analysis of the Consolidated XAI Discovery

The synthesized interpretability parameters recorded in Table 7.8 establish high algorithmic accountability for deep multi-scale environmental tracking. In contemporary air quality informatics literature, high-performing model fusions are frequently limited by severe interpretability vulnerabilities; while advanced architectures optimize statistical error drops, they routinely operate as unverified "black boxes." Even when explainability layers are added, researchers traditionally deploy a single tool (either SHAP or LIME in isolation), leaving the model exposed to single-explainer mathematical biases or unvalidated localized artifacts. This research completely overcomes the limitation by establishing a dual-layered,cross-verified verification pipeline where SHAP and LIME are forced to check the identical model configuration.

The data transitions mapped in the consolidated matrix shows that the framework has learned authentic regional atmospheric chemistry rather than simple historical noise.Both the explainability models supports the finding by each of the model.

Chapter 8

Conclusion

8.1 Summary of Research Accomplishments

This work has successfully showed the validity of the SHAP-Weighted Adaptive Ensemble (SWAE) architecture as a strong computational engine for urban air quality informatics.The framework ha successfully resolved the historic trade-off between predictive optimization and machine learning transparency.Tested across six urban cities using Central Pollution Control Board (CPCB) telemetry from 2015 to 2020,the system proved resilient against non-linear pollutant dispersion .varying patterns and sudden concentration shocks.

Major three findings has been validated by this work:

- **Substantial Error Reduction:** In industry heavy , high-traffic regions like Delhi,the SWAE optimization model reduced residual tracking errors by up to 68.93% compared to standalone sequential baselines.
- **Dynamic Structural Fallback:** In complex coastal domains like Chennai,where macro-seasonal maritime land-sea breeze reversals ,the models dynamically adapts the decision making, maintaining model stability.
- **Closed-Loop Explainability Validation:**By implementation and cross validation of two simultaneously working models strongly justifies the decision making of the optimized framework .

8.2 Future Work

To build upon the structural milestones established by the SWAE network, three future research directions are proposed:

1. **Multimodal Spatial Data Integration:** Incorporating satellite-derived data grids.
2. **Online Stream Learning Scalability:** Transitioning the ensemble optimization layer into a continuous,online learning framework capable of adapting feature weights dynamically as live streams of atmospheric metrics flow through edge nodes.

References

[1] N. K. Arora and I. Mishra, "United nations sustainable development goals 2030 and environmental sustainability: race against time," *Environmental Sustainability*, vol. 2, no. 4, pp. 339–342, 2019.

[2] P. R. Prashant Rajput, G. K. Singh, and T. G. Tarun Gupta, "Indices used for assessment of air quality.," in *Air pollution: sources, impacts and controls*, pp. 96–112, CAB International Wallingford UK, 2019.

[3] IQAir Team, "5th annual world air quality report: Region & city pm2.5 ranking," technical report, IQAir Air Quality Science Department, Goldach, Switzerland, 2023.

[4] K. K. Sidhu, H. Balogun, and K. O. Oseni, "Predictive modelling of air quality index (aqi) across diverse cities and states of india using machine learning: Investigating the influence of punjab's stubble burning on aqi variability," 2024.

[5] A. M. Salih, Z. Raisi-Estabragh, I. B. Galazzo, P. Radeva, S. E. Petersen, K. Lekadir, and G. Menegaz, "A perspective on explainable artificial intelligence methods: Shap and lime," *Advanced Intelligent Systems*, vol. 7, no. 1, 2024.

[6] K. Kumar and B. P. Pande, "Air pollution prediction with machine learning: a case study of indian cities," *International Journal of Environmental Science and Technology*, vol. 20, pp. 5333–5348, May 2023.

[7] A. Dubey, M. Kumar, S. Mishra, and S. Targetti, "Transforming air pollution management in india with ai and machine learning technologies," *Scientific Reports*, vol. 14, no. 1, p. 6795, 2024.

[8] A. Masood and K. Ahmad, "Data-driven predictive modeling of $PM_{2.5}$ concentrations using machine learning and deep learning techniques: a case study of Delhi, India," *Environmental Monitoring and Assessment*, vol. 195, p. 60, January 2023.

[9] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[10] M. T. Ribeiro, S. Singh, and C. Guestrin, ""Why should I trust you?": Explaining the predictions of any classifier," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135–1144, 2016.

[11] Central Pollution Control Board, India, "Air quality data in India (2015–2020)." Kaggle Dataset, uploaded by Rohan Rao, 2020. Originally compiled from CPCB monitoring stations.

[12] S. K. Natarajan, P. Shanmurthy, D. Arockiam, B. Balusamy, and S. Selvarajan, "Optimized machine learning model for air quality index prediction in major cities in india," *Scientific Reports*, vol. 14, p. 6795, March 2024.

[13] G. Ravindiran *et al.*, "Ensemble stacking of machine learning models for air quality index prediction in Hyderabad, India," *iScience*, 2025.

[14] A. Yenkikar *et al.*, "Explainable forecasting of air quality index using hybrid Random Forest and ARIMA model," *MethodsX*, 2025.

[15] M. Hussain *et al.*, "Deep learning-based AQI forecasting: CNN-LSTM with SHAP, LIME, and PDP explainability," *Discover Applied Sciences*, 2025.

[16] M. Qasim *et al.*, "Scalable AI-driven AQI forecasting with SHAP and LIME interpretability," *Discover Atmosphere*, 2025.

[17] G. Ravindiran *et al.*, "Air quality prediction by machine learning models: A predictive study on the Indian coastal city of Visakhapatnam," *Chemosphere*, vol. 338, p. 139518, 2023.

[18] A. Bayramoglu, C. Yilmaz, and M. Demirel, "Seasonally stratified ensemble arrays with game-theoretic attributions for urban air quality forecasting in istanbul," *Atmospheric Pollution Research*, vol. 17, no. 2, pp. 101420–101435, 2026.

[19] L. Zhao *et al.*, "SHAP-based feature importance for multi-pollutant AQI forecasting using XGBoost and LSTM," *Sustainability*, 2024.

[20] M.-K. Kim, J.-Y. Kim, T.-H. Kim, and S.-U. Choi, "Explainable ai-based sequential deep learning architectures for domain-specific prediction and feature importance validation," *IEEE Access*, vol. 11, pp. 43210–43225, 2023.